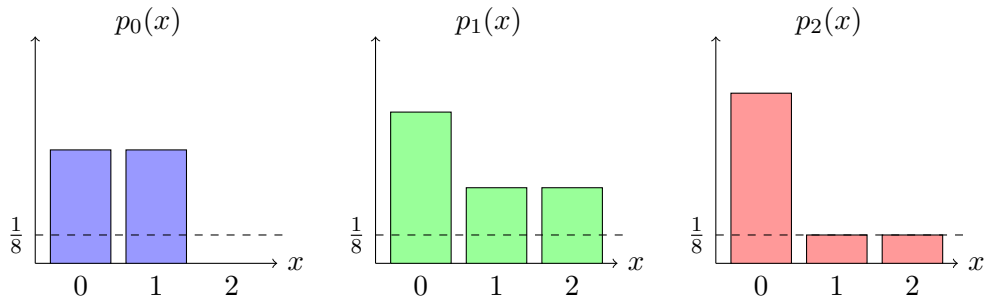


Quiz 7: Quantum Compression and Entropy

Question 1. Consider the probability distributions.



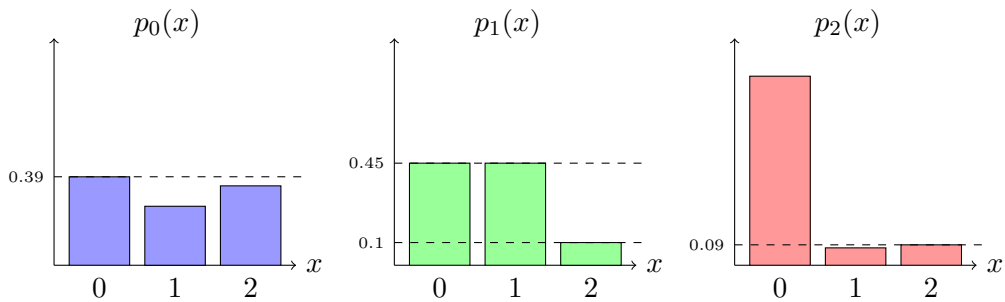
Which of these has the smallest optimal zero-error compression $C^0(p_i)$?

- (a) p_0 , (b) p_1 , (c) p_2 ,

Question 2. Which of the probability distributions from the previous question has the smallest optimal compression with error $1/4$, that is, $C^{1/4}(p_i)$?

- (a) p_0 , (b) p_1 , (c) p_2 ,

Question 3. Consider the following probability distributions.



Which one has the largest Shannon entropy $H(p_i)$?

- (a) p_0 , (b) p_1 , (c) p_2 ,

Question 4. Let p_X be a probability distribution. Which of the following statements about the typical sets $T_{n,\varepsilon}(p_X)$ are true in general?

- (a) $T_{n,\varepsilon}(p_X)$ contains the most likely outcomes for the variables $\mathbf{X}^n$.

(b) $T_{n,\varepsilon}(p_X)$ gets larger as $n \rightarrow \infty$.

(c) The probability that a sample of $\mathbf{X}^n$ is in $T_{n,\varepsilon}(p_X)$ is equal to one.

Question 5. Suppose Alice wants to send Bob a message consisting of n bits. Each bit is independently 0 with probability $3/4$, and 1 with probability $1/4$. How many bits does she need to send in the limit of large n ?

(a) $\frac{n}{4}(6 - 3 \log 3)$, (b) $n \left(2 - \frac{3}{4} \log 3 \right)$, (c) $2n$, (d) n ,